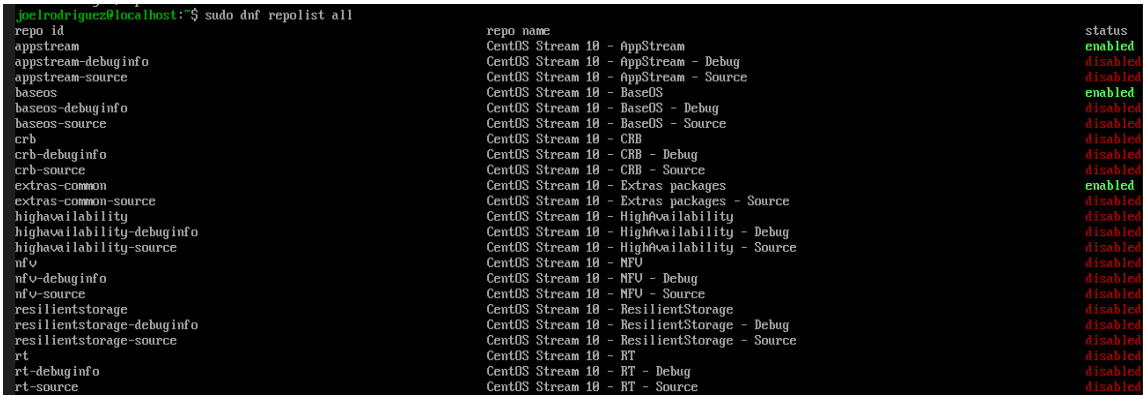
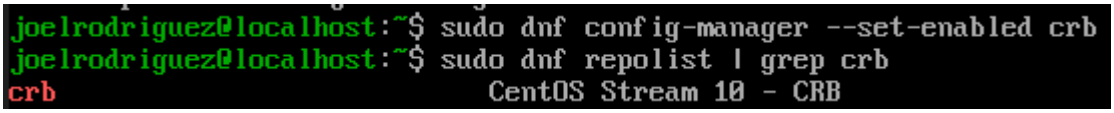
Joel Rodriguez
CIS-245
Fall 2026
Documentation

CENTOS

1. Check Repositories that are included by default
    a. Use *sudo dnf repolist all* command to show all the current repos enabled and diasbled



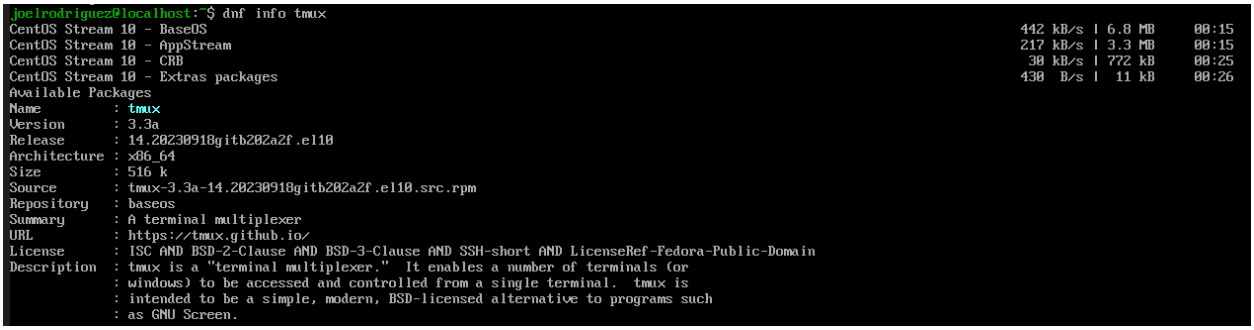
    b. Add a repository
        1. Official – Run this command to enable the CRB repo: *sudo dnf config-manager –set-enabled crb*



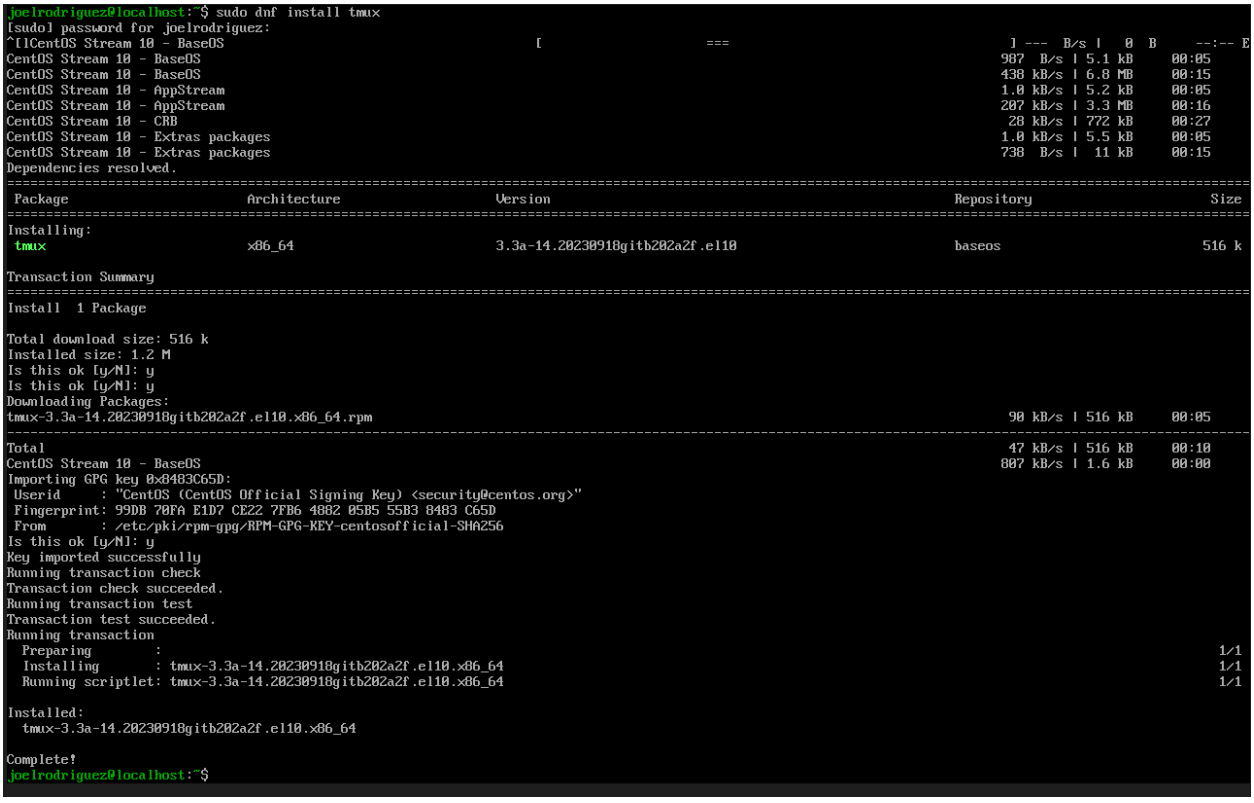
        2. Verify the repo is enabled with: *sudo dnf repolist*



2. Install Tmux
    a. We're going to check if DNF can find Tmux with this command: *dnf info tmux*



    b. Since the package was found now we're going to install it: *sudo dnf install tmux*
    c. It will ask if the file size is ok? Enter y to proceed



    d. Verify that tmux is installed with: *tmux – V*



3. Install emacs
    a. We're going to check if DNF can find emacs: *dnf info emacs*

```
joelrodriguez@localhost:~$ dnf info emacs
Last metadata expiration check: 0:17:53 ago on Fri 02 Oct 2026 03:00:40 PM EDT.
Available Packages
Name         : emacs
Epoch        : 1
Version      : 29.4
Release      : 13.el10
Architecture : x86_64
Size         : 3.7 M
Source       : emacs-29.4-13.el10.src.rpm
Repository   : appstream
Summary      : GNU Emacs text editor
URL          : http://www.gnu.org/software/emacs/
License      : GPL-3.0-or-later AND CC0-1.0
Description  : Emacs is a powerful, customizable, self-documenting, modeless text
             : editor. Emacs contains special code editing features, a scripting
             : language (elisp), and the capability to read mail, news, and more
             : without leaving the editor.
             :
             : This package provides an emacs binary with support for Wayland, using the
             : GTK toolkit.
```

    b. Since the package was found now we're going to install it: *sudo dnf install emacs*
    c. It will ask if the file size is ok? Enter y to proceed
    d. Verify that tmux is installed with: *emacs --version*

```
joelrodriguez@localhost:~$ emacs --version
GNU Emacs 29.4
Copyright (C) 2024 Free Software Foundation, Inc.
GNU Emacs comes with ABSOLUTELY NO WARRANTY.
You may redistribute copies of GNU Emacs
under the terms of the GNU General Public License.
For more information about these matters, see the file named COPYING.
```

4. Install Fail2Ban
    a. We're going to check if DNF can find fail2ban: *dnf info fail2ban*

```
joelrodriguez@localhost:~$ dnf info fail2ban
Last metadata expiration check: 0:28:54 ago on Fri 02 Oct 2026 03:00:40 PM EDT.
Error: No matching Packages to list
```

    b. DNF won't find the packages because EPEL isn't enabled.
    c. Let's verify that EPEL is enabled: *dnf repolist | grep epel*

d. If there's no output, we're going to install EPEL into the OS: *sudo dnf install epel-release*
e. Verify that EPEL was installed: *dnf repolist | grep epel*

```
joelrodriguez@localhost:~$ dnf repolist | grep epel
epel                    Extra Packages for Enterprise Linux 10 - x86_64
```

f. DNF should find fail2ban now: *dnf info fail2ban*

```
joelrodriguez@localhost:~$ dnf info fail2ban
Extra Packages for Enterprise Linux 10 - x86_64                 390 kB/s | 7.2 MB     00:19
Last metadata expiration check: 0:00:03 ago on Fri 02 Oct 2026 03:45:03 PM EDT.
Available Packages
Name         : fail2ban
Version      : 1.1.0
Release      : 6.el10_0
Architecture : noarch
Size         : 9.4 k
Source       : fail2ban-1.1.0-6.el10_0.src.rpm
Repository   : epel
Summary      : Daemon to ban hosts that cause multiple authentication errors
URL          : https://www.fail2ban.org
License      : GPL-2.0-or-later
Description  : Fail2Ban scans log files and bans IP addresses that makes too many password
             : failures. It updates firewall rules to reject the IP address. These rules can
             : be defined by the user. Fail2Ban can read multiple log files such as sshd or
             : Apache web server ones.
             :
             : Fail2Ban is able to reduce the rate of incorrect authentications attempts
             : however it cannot eliminate the risk that weak authentication presents.
             : Configure services to use only two factor or public/private authentication
             : mechanisms if you really want to protect services.
             :
             : This is a meta-package that will install the default configuration.  Other
             : sub-packages are available to install support for other actions and
             : configurations.
```

g. Now we're going to install fail2ban: *sudo dnf install fail2ban*
h. It will ask if the file size is ok? Enter y to proceed
i. Verify that fail2ban is installed: *fail2ban-client –version*

```
joelrodriguez@localhost:~$ fail2ban-client --version
Fail2Ban v1.1.0
```

j. Make sure to make changes to the settings needed to get it working.
    1. Let's check if the service for fail2ban is running:
    sudo systemctl status fail2ban

```
joelrodriguez@localhost:~$ sudo systemctl status fail2ban
● fail2ban.service - Fail2Ban Service
     Loaded: loaded (/usr/lib/systemd/system/fail2ban.service; disabled; preset: disabled)
     Active: inactive (dead)
       Docs: man:fail2ban(1)
```

    2. If it's inactive we're going to enable it: *sudo systemctl enable –now fail2ban*

```
joelrodriguez@localhost:~$ sudo systemctl enable --now fail2ban
Created symlink '/etc/systemd/system/multi-user.target.wants/fail2ban.service' → '/usr/lib/systemd/system/fail2ban.service'.
```

    3. Verify that it's active: *sudo systemctl status fail2ban*

```
joelrodriguez@localhost:~$ sudo systemctl status fail2ban
● fail2ban.service - Fail2Ban Service
     Loaded: loaded (/usr/lib/systemd/system/fail2ban.service; enabled; preset: disabled)
     Active: active (running) since Fri 2026-10-02 16:01:11 EDT; 43s ago
 Invocation: c14e68b00fe64d368ecf63b1cb5270d2
       Docs: man:fail2ban(1)
    Process: 4128 ExecStartPre=/bin/mkdir -p /run/fail2ban (code=exited, status=0/SUCCESS)
   Main PID: 4129 (fail2ban-server)
      Tasks: 3 (limit: 22981)
     Memory: 11.9M (peak: 13.9M)
        CPU: 95ms
     CGroup: /system.slice/fail2ban.service
             └─4129 /usr/bin/python3 -sP /usr/bin/fail2ban-server -xf start

Oct 02 16:01:11 localhost.localdomain systemd[1]: Starting fail2ban.service - Fail2Ban Service...
Oct 02 16:01:11 localhost.localdomain systemd[1]: Started fail2ban.service - Fail2Ban Service.
Oct 02 16:01:11 localhost.localdomain fail2ban-server[4129]: Server ready
```

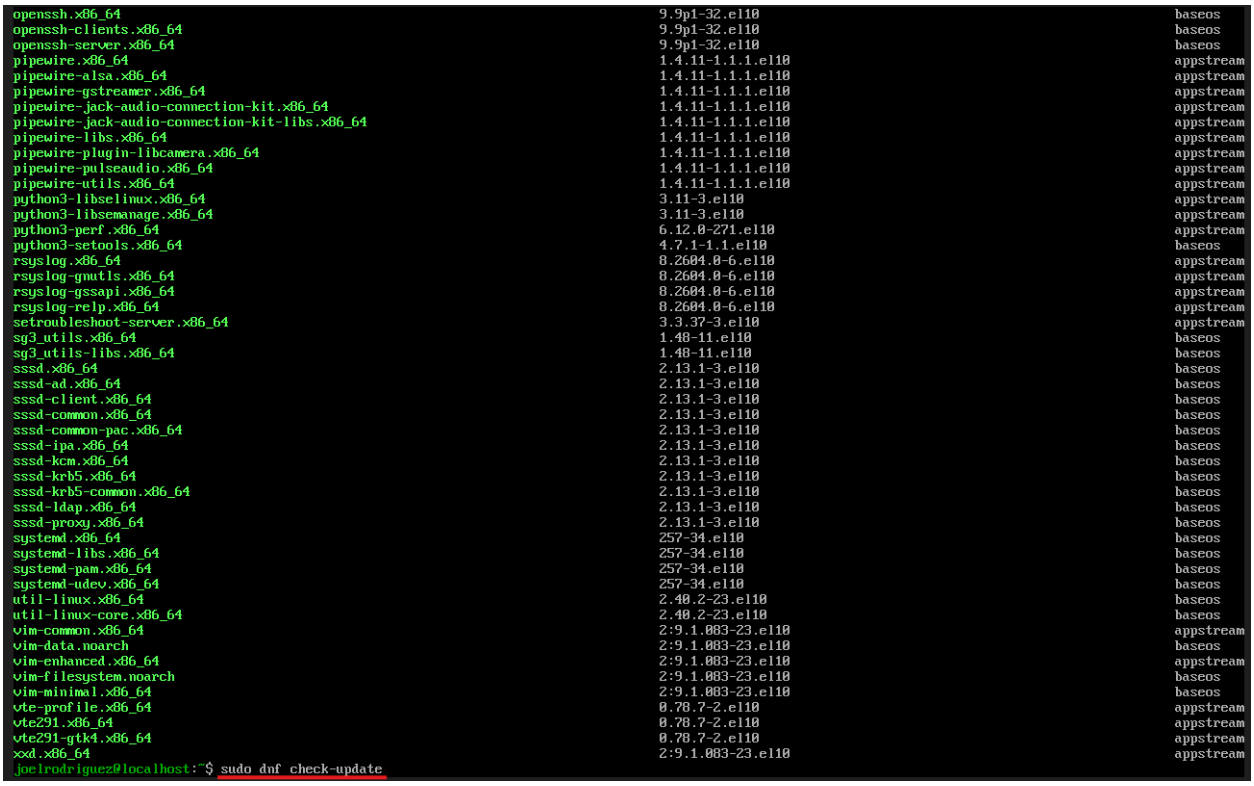
5. Install cowsay
    a. Run this command: *sudo dnf install cowsay*
    b. It will ask if the file size is ok? Enter y to proceed
    c. Verify it's installed: *cowsay –version*

```
joelrodriguez@localhost:~$ cowsay --version
/usr/bin/cowsay version 3.8.4 calling Getopt::Std::getopts (version 1.14),
running under Perl version 5.40.2.
```

6. Install lolcat
    a. Run this command: *sudo dnf install lolcat*
    b. It will ask if the file size is ok? Enter y to proceed
    c. Verify it's installed:

```
joelrodriguez@localhost:~$ lolcat --version
lolcat version 1.5, (c) 2024 jaseg
```

7. Update the server
    a. Check if there are any updates available: *sudo dnf check-update*



    b. We're going to update everything that's available: *sudo dnf upgrade*

c. It's going to show a transaction summary and ask if the download size is ok? Enter y to proceed.

```
Transaction Summary
================================================================================
Install    5 Packages
Upgrade  115 Packages

Total download size: 306 M
Is this ok [y/N]:
```

d. All the updates will install, and you will get a "Complete!" for a confirmation.

8. Update VI to VIM
   a. I checked if we had any versions of vi or vim installed: *which vi/vim*

```
joelrodriguez@localhost:~$ which vi
/usr/bin/vi
joelrodriguez@localhost:~$ which vim
/usr/bin/vim
```

   b. I checked the installed vim packages: rpm -q vim-minimal/enhanced

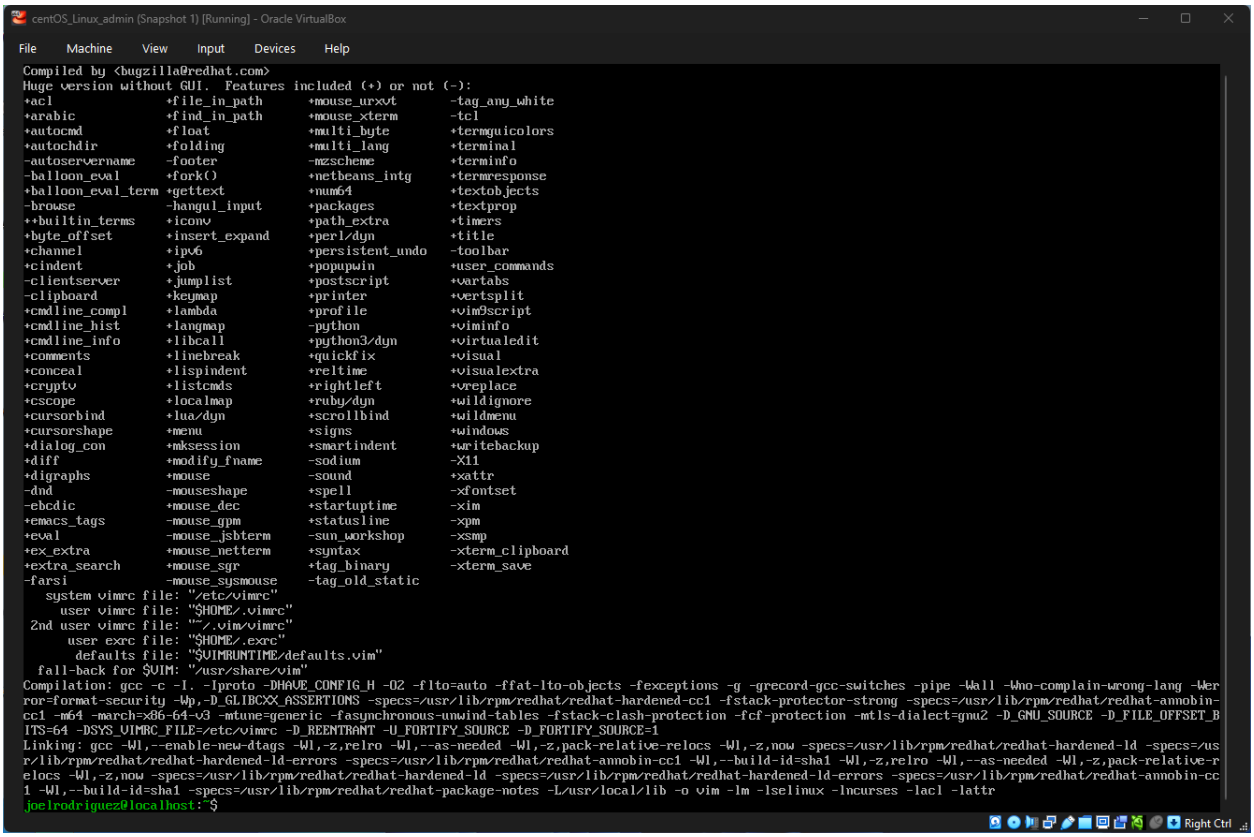
```
joelrodriguez@localhost:~$ rpm -q vim-minimal
vim-minimal-9.1.083-23.el10.x86_64
joelrodriguez@localhost:~$ rpm -q vim-enhanced
vim-enhanced-9.1.083-23.el10.x86_64
```

   this confirms that both vim packages are already installed.

   c. Make sure the full version of vim is installed or updated: *sudo dnf install vim-enhanced*

```
joelrodriguez@localhost:~$ sudo dnf install vim-enhanced
[sudo] password for joelrodriguez:
CentOS Stream 10 - BaseOS
CentOS Stream 10 - AppStream
CentOS Stream 10 - CRB
CentOS Stream 10 - Extras packages
Package vim-enhanced-2:9.1.083-23.el10.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
```

   d. Verify vim after the update: *vim --version*

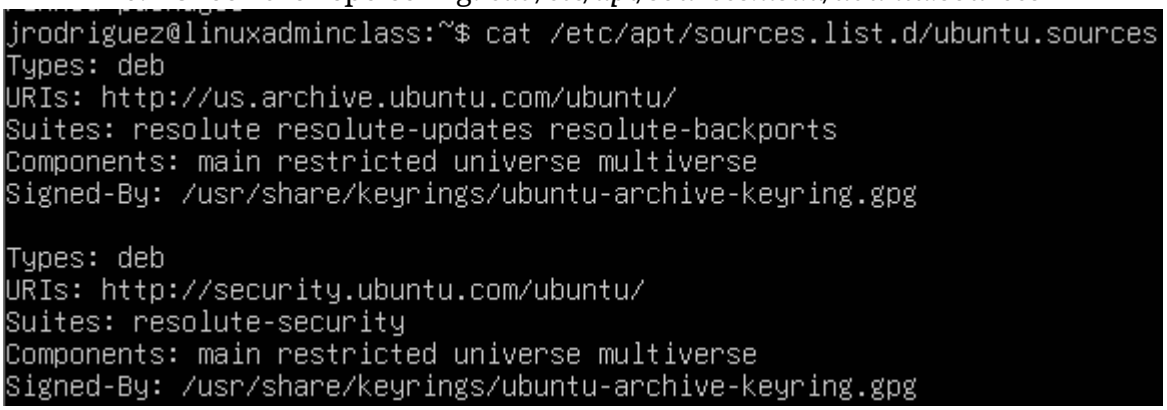


Ubuntu

9. Check Repositories that are included by default
    a. First, we're going to check which version of Ubuntu was installed: *cat /etc/os-release*

```
jrodriguez@linuxadminclass:~$ cat /etc/os-release
PRETTY_NAME="Ubuntu 26.04.1 LTS"
NAME="Ubuntu"
VERSION_ID="26.04"
VERSION="26.04.1 LTS (Resolute Raccoon)"
VERSION_CODENAME=resolute
ID=ubuntu
ID_LIKE=debian
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"
UBUNTU_CODENAME=resolute
LOGO=ubuntu-logo
```

b. Check the repo config: *cat /etc/apt/sources.list.d/ubuntu.sources*

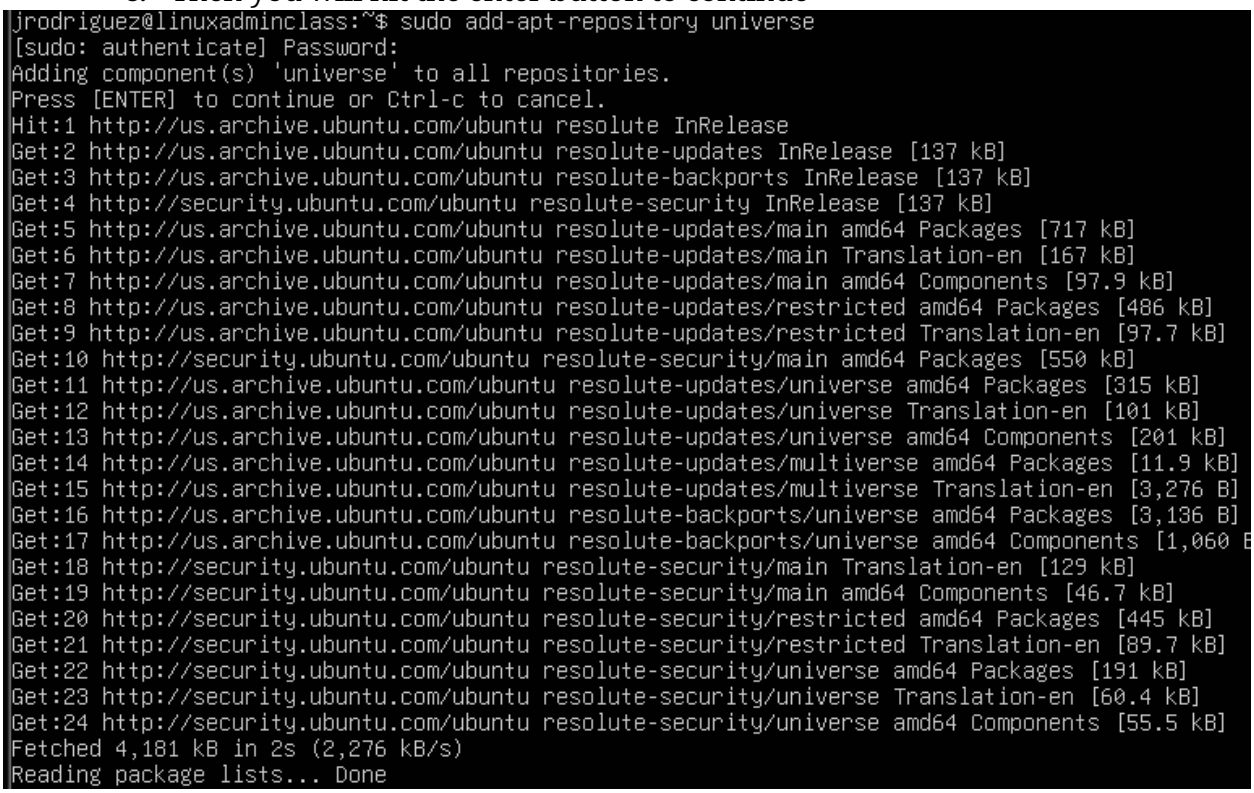


c. Verify the repos APT is using: *apt policy*

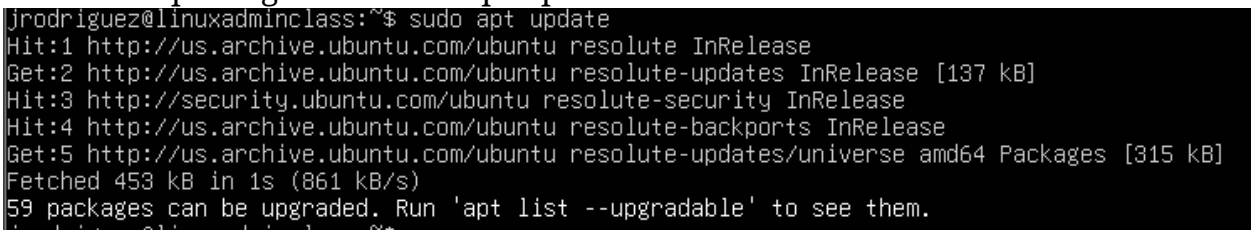
```
jrodriguez@linuxadminclass:~$ apt policy
Package files:
 100 /var/lib/dpkg/status
     release a=now
 500 http://security.ubuntu.com/ubuntu resolute-security/multiverse amd64 Packages
     release v=26.04,o=Ubuntu,a=resolute-security,n=resolute,l=Ubuntu,c=multiverse,b=amd64
     origin security.ubuntu.com
 500 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Packages
     release v=26.04,o=Ubuntu,a=resolute-security,n=resolute,l=Ubuntu,c=universe,b=amd64
     origin security.ubuntu.com
 500 http://security.ubuntu.com/ubuntu resolute-security/restricted amd64 Packages
     release v=26.04,o=Ubuntu,a=resolute-security,n=resolute,l=Ubuntu,c=restricted,b=amd64
     origin security.ubuntu.com
 500 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages
     release v=26.04,o=Ubuntu,a=resolute-security,n=resolute,l=Ubuntu,c=main,b=amd64
     origin security.ubuntu.com
 100 http://us.archive.ubuntu.com/ubuntu resolute-backports/universe amd64 Packages
     release v=26.04,o=Ubuntu,a=resolute-backports,n=resolute,l=Ubuntu,c=universe,b=amd64
     origin us.archive.ubuntu.com
 500 http://us.archive.ubuntu.com/ubuntu resolute-updates/multiverse amd64 Packages
     release v=26.04,o=Ubuntu,a=resolute-updates,n=resolute,l=Ubuntu,c=multiverse,b=amd64
     origin us.archive.ubuntu.com
 500 http://us.archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages
     release v=26.04,o=Ubuntu,a=resolute-updates,n=resolute,l=Ubuntu,c=universe,b=amd64
     origin us.archive.ubuntu.com
 500 http://us.archive.ubuntu.com/ubuntu resolute-updates/restricted amd64 Packages
     release v=26.04,o=Ubuntu,a=resolute-updates,n=resolute,l=Ubuntu,c=restricted,b=amd64
     origin us.archive.ubuntu.com
 500 http://us.archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages
     release v=26.04,o=Ubuntu,a=resolute-updates,n=resolute,l=Ubuntu,c=main,b=amd64
     origin us.archive.ubuntu.com
 500 http://us.archive.ubuntu.com/ubuntu resolute/multiverse amd64 Packages
     release v=26.04,o=Ubuntu,a=resolute,n=resolute,l=Ubuntu,c=multiverse,b=amd64
     origin us.archive.ubuntu.com
 500 http://us.archive.ubuntu.com/ubuntu resolute/universe amd64 Packages
     release v=26.04,o=Ubuntu,a=resolute,n=resolute,l=Ubuntu,c=universe,b=amd64
     origin us.archive.ubuntu.com
 500 http://us.archive.ubuntu.com/ubuntu resolute/restricted amd64 Packages
     release v=26.04,o=Ubuntu,a=resolute,n=resolute,l=Ubuntu,c=restricted,b=amd64
     origin us.archive.ubuntu.com
 500 http://us.archive.ubuntu.com/ubuntu resolute/main amd64 Packages
     release v=26.04,o=Ubuntu,a=resolute,n=resolute,l=Ubuntu,c=main,b=amd64
     origin us.archive.ubuntu.com
Pinned packages:
```

10. Add an official repository

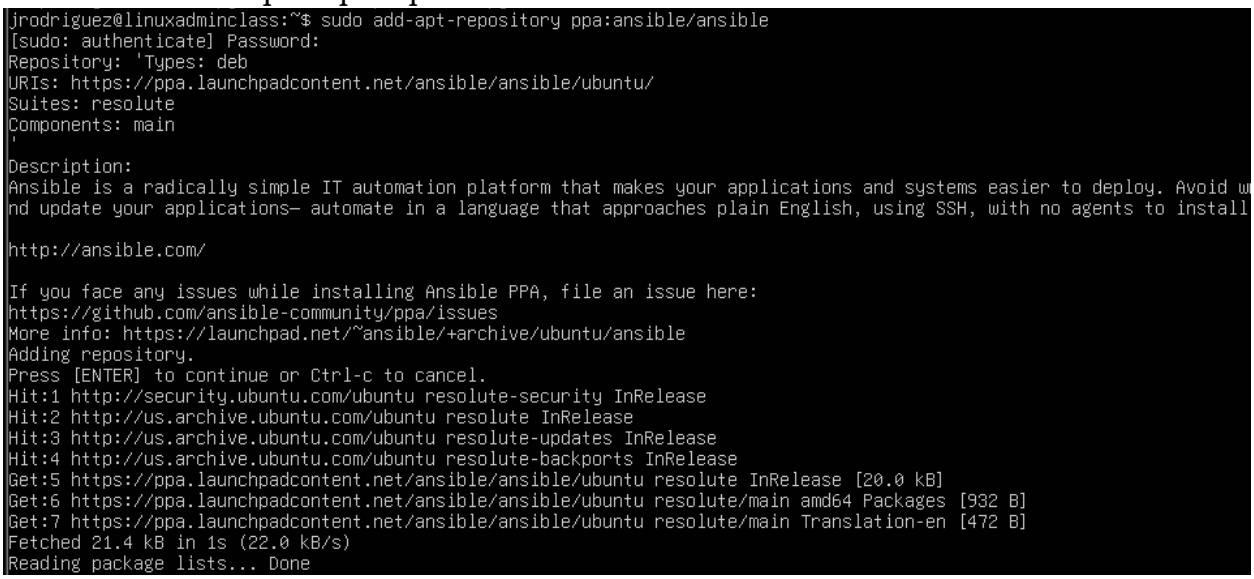
a. I used the following command to enable the official Universe repo: *sudo add-apt-repository universe*
b. You will enter your password
c. Then you will hit the enter button to continue



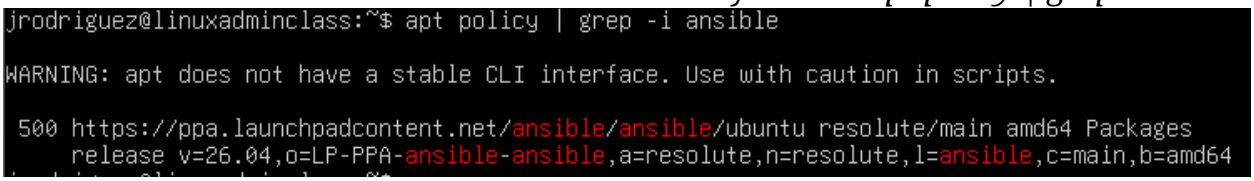
d. Now that the official repo is enabled I am going to update the APT's package info: sudo apt update



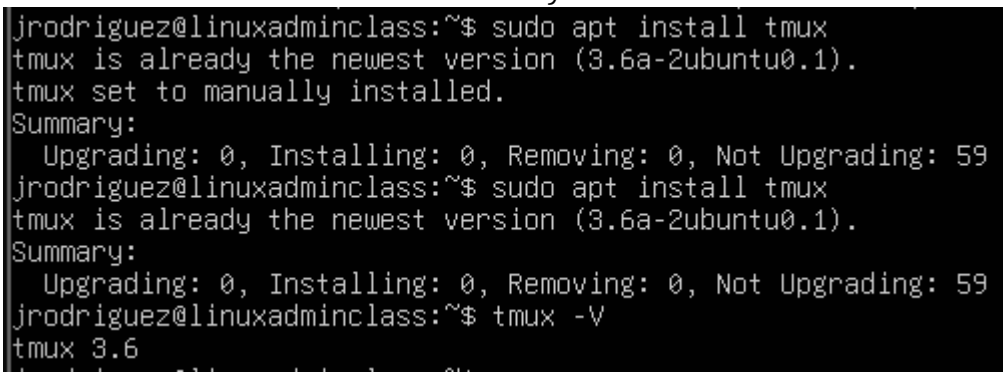
11. Add a PPA repository
    a. Let's add the Ansible PPA: sudo add-apt-repository ppa:ansible/ansible
    b. You will enter your password
    c. When prompted press enter to continue



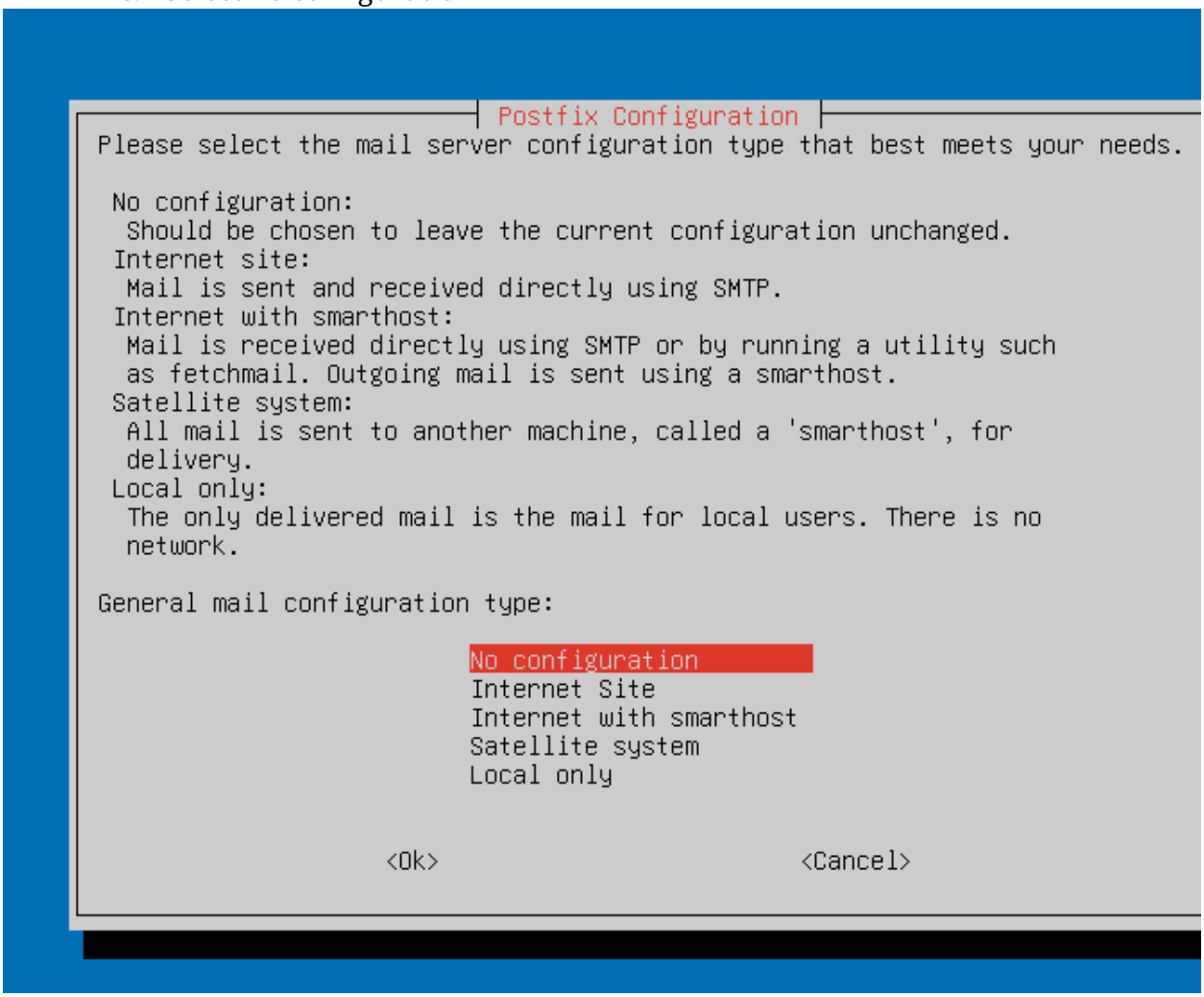
    d. Update APT again: *sudo apt update*
    e. Verified that the PPA was successfully added: *apt policy | grep -I ansible*



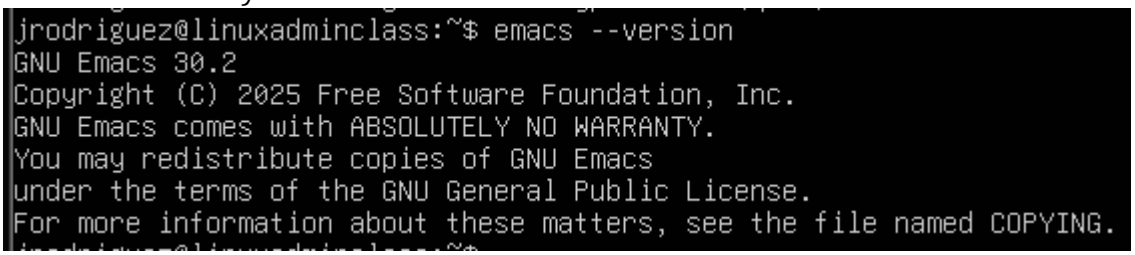
12. Install Tmux
    a. I attempted to install Tmux using: *sudo apt install tmux*
    b. I verified Tmux was already installed: tmux -V



13. Install emacs
    a. I installed emacs using: *sudo apt install emacs*
    b. Enter y to continue
    c. Select no configuration



    d. Verify emacs was installed: emacs –version



14. Install Fail2Ban
    a. I installed fail2ban using: *sudo apt install fail2ban*
    b. Enter Y when prompted
    c. Verify that the installation completed: *fail2ban-client --version*
    d. Make sure to make changes to the settings needed to get it working.
        1. We're going to check if the service is running*: sudo systemctl status fail2ban*

```
jrodriguez@linuxadminclass:~$ sudo systemctl status fail2ban
● fail2ban.service - Fail2Ban Service
     Loaded: loaded (/usr/lib/systemd/system/fail2ban.service; enabled; preset: enabled)
     Active: active (running) since Mon 2026-10-05 03:08:35 UTC; 8min ago
 Invocation: 8e2bf5bfa5d54985beb60d8110a3c325
       Docs: man:fail2ban(1)
   Main PID: 5169 (fail2ban-server)
      Tasks: 5 (limit: 3993)
     Memory: 15.4M (peak: 15.7M)
        CPU: 985ms
     CGroup: /system.slice/fail2ban.service
             └─5169 /usr/bin/python3 /usr/bin/fail2ban-server -xf start

Oct 05 03:08:35 linuxadminclass systemd[1]: Started fail2ban.service - Fail2Ban Service.
Oct 05 03:08:35 linuxadminclass fail2ban-server[5169]: Server ready
```

15. Install cowsay
    a. Installed cowsay using: *sudo apt install cowsay*
    b. Verified the installation completed using: *apt list –installed | grep cowsay*

```
jrodriguez@linuxadminclass:~$ apt list --installed | grep cowsay

WARNING: apt does not have a stable CLI interface. Use with caution in scripts.

cowsay/resolute,now 3.03+dfsg2-8build1 all [installed]
```

16. Install lolcat
    a. Installed lol using: *sudo apt install lolcat*
    b. Enter y when prompted to continue.
    c. Verify the installation was completed: lolcat --version

```
jrodriguez@linuxadminclass:~$ lolcat --version
lolcat 100.0.1 (c)2011 moe@busyloop.net
```

17. Update the server
    a. To refresh the package list I ran: *sudo apt update*
    b. Next, I checked what was able to be updated: apt list –upgradeable
    c. Install the updates: *sudo apt upgrade*
    d. Enter y when prompted
    e. Verify that the Server is updated: *sudo apt update*
    f. It should say all packages are up to date

```
jrodriguez@linuxadminclass:~$ sudo apt update
Hit:1 http://us.archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://us.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:3 http://us.archive.ubuntu.com/ubuntu resolute-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu resolute-security InRelease
Hit:5 https://ppa.launchpadcontent.net/ansible/ansible/ubuntu resolute InRelease
All packages are up to date.
```

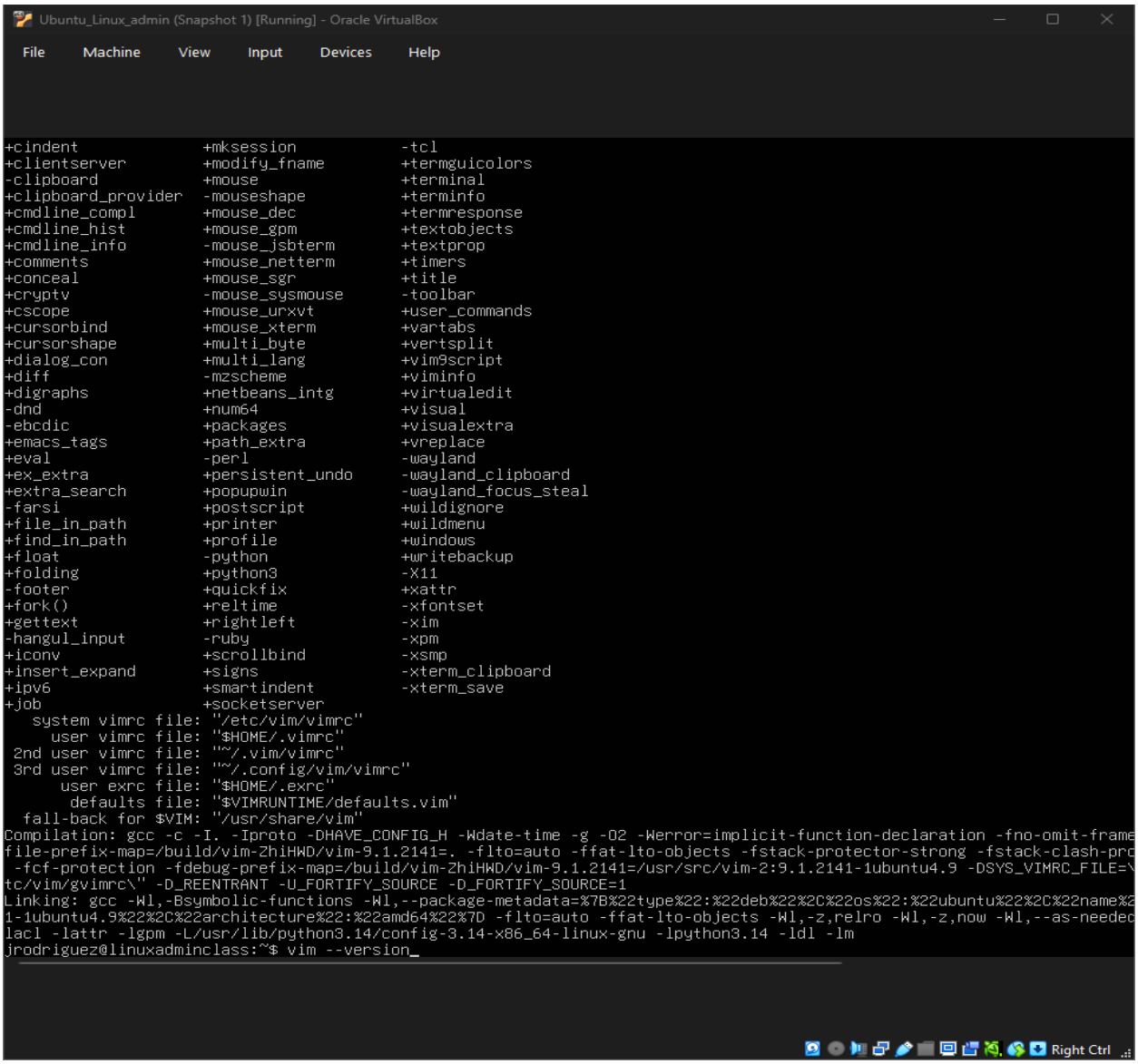
18. Update VI to VIM
    a. First, I checked whether VI and VIM were available by running: *which vi/vim*

```
jrodriguez@linuxadminclass:~$ which vi
/usr/bin/vi
jrodriguez@linuxadminclass:~$ which vim
/usr/bin/vim
```

    b. Then, I checked the installed vim packages: *dpkg -l | grep vim*

```
jrodriguez@linuxadminclass:~$ dpkg -l | grep vim
ii  vim                                  2:9.1.2141-1ubuntu4.9                   amd64        Vi IMproved - enhan
ii  vim-common                           2:9.1.2141-1ubuntu4.9                   all          Vi IMproved - Commo
ii  vim-runtime                          2:9.1.2141-1ubuntu4.9                   all          Vi IMproved - Runti
ii  vim-tiny                             2:9.1.2141-1ubuntu4.9                   amd64        Vi IMproved - enhan
```

    c. Verified that vim was already installed by using: *vim --version*


Ubuntu_Linux_admin (Snapshot 1) [Running] - Oracle VirtualBox
File Machine View Input Devices Help
+cindent +mksession -tcl
+clientserver +modify_fname +termguicolors
-clipboard +mouse +terminal
+clipboard_provider -mouseshape +terminfo
+cmdline_compl +mouse_dec +termresponse
+cmdline_hist +mouse_gpm +textobjects
+cmdline_info -mouse_jsbterm +textprop
+comments +mouse_netterm +timers
+conceal +mouse_sgr +title
+cryptv -mouse_sysmouse -toolbar
+cscope +mouse_urxvt +user_commands
+cursorbind +mouse_xterm +vartabs
+cursorshape +multi_byte +vertsplit
+dialog_con +multi_lang +vim9script
+diff -mzscheme +viminfo
+digraphs +netbeans_intg +virtualedit
-dnd +num64 +visual
-ebcdic +packages +visualextra
+emacs_tags +path_extra +vreplace
+eval -perl -wayland
+ex_extra +persistent_undo -wayland_clipboard
+extra_search +popupwin -wayland_focus_steal
-farsi +postscript +wildignore
+file_in_path +printer +wildmenu
+find_in_path +profile +windows
+float -python +writebackup
+folding +python3 -X11
-footer +quickfix +xattr
+fork() +reltime -xfontset
+gettext +rightleft -xim
-hangul_input -ruby -xpm
+iconv +scrollbind -xsmp
+insert_expand +signs -xterm_clipboard
+ipv6 +smartindent -xterm_save
+job +socketserver
system vimrc file: "/etc/vim/vimrc"
user vimrc file: "$HOME/.vimrc"
2nd user vimrc file: "~/.vim/vimrc"
3rd user vimrc file: "~/.config/vim/vimrc"
user exrc file: "$HOME/.exrc"
defaults file: "$VIMRUNTIME/defaults.vim"
fall-back for $VIM: "/usr/share/vim"
Compilation: gcc -c -I. -Iproto -DHAVE_CONFIG_H -Wdate-time -g -O2 -Werror=implicit-function-declaration -fno-omit-frame
file-prefix-map=/build/vim-ZhiHWD/vim-9.1.2141=. -flto=auto -ffat-lto-objects -fstack-protector-strong -fstack-clash-pro
-fcf-protection -fdebug-prefix-map=/build/vim-ZhiHWD/vim-9.1.2141=/usr/src/vim-2:9.1.2141-1ubuntu4.9 -DSYS_VIMRC_FILE=\
tc/vim/gvimrc\" -D_REENTRANT -U_FORTIFY_SOURCE -D_FORTIFY_SOURCE=1
Linking: gcc -Wl,-Bsymbolic-functions -Wl,--package-metadata=%7B%22type%22:%22deb%22%2C%22os%22:%22ubuntu%22%2C%22name%2
1-1ubuntu4.9%22%2C%22architecture%22:%22amd64%22%7D -flto=auto -ffat-lto-objects -Wl,-z,relro -Wl,-z,now -Wl,--as-needed
lacl -lattr -lgpm -L/usr/lib/python3.14/config-3.14-x86_64-linux-gnu -lpython3.14 -ldl -lm
jrodriguez@linuxadminclass:~$ vim --version_
Right Ctrl